

DAXHIEL PERUGORRIA RUCIEL

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EDUCATION

University of Texas at Austin – Austin, TX

May 2029

Bachelor of Science in Computer Science

GPA: 3.4

RELEVANT COURSEWORK

Computer Organization and Architecture, Linear Algebra, Probability and Statistics Data Structures, Calculus, Discrete Math

PROJECTS

Kalshi Arbitrage Bot | Python, REST

- **Developed a Python-based arbitrage detection engine** that scans Kalshi prediction markets using REST API integration with RSA-signature authentication, identifying risk-free profit opportunities by analyzing bundle arbitrage patterns across market outcomes.
- **Implemented automated market analysis algorithms** processing 1,000+ events through paginated API calls with intelligent rate limiting, filtering active markets and calculating arbitrage spreads with precision to detect opportunities where outcome price sums deviate from theoretical equilibrium.
- **Engineered a modular architecture with custom data models** and object-oriented design patterns, including API client wrapper with cryptographic request signing, datetime parsing, and validation logic to ensure market efficiency across production and demo trading environments

Fake-or-Not | Python, FastAPI, React

- **Built an end-to-end fake-news classification pipeline** using Jupyter notebooks for data exploration and feature engineering with data libraries, and exported the trained model as an ONNX artifact for portable inference.
- **Implemented a production-ready model** serving backend exposing a REST API for real-time predictions, including request validation, model versioning, and batch prediction endpoints to support both online and offline workflows.
- **Developed a responsive single-page frontend** in React (Vite) with fetch-based integration to the REST /predict endpoint, styled with modern CSS and maintained with ESLint for code quality.

Kwii Programming Language | Java, GitHub

- **Designed and implemented an interpreted programming language** from scratch, inspired by Crafting Interpreters and using a recursive descent parser, abstract syntax tree evaluator, and runtime environment supporting variables, control flow, and functions.
- **Applied compiler theory concepts** (lexical analysis, parsing, runtime environments) to create a modular and extensible interpreter.

EXPERIENCE

Austin Spanish First – Austin, TX

November 2020 - Current

Co-Director

- **Directed a 10+ member technical team**, delegating tasks and coordinating efforts to deliver seamless audio/visual experiences for 400+ weekly attendees.
- **Led the redesign and optimization** of audio/streaming infrastructure, improving broadcast quality and increasing online viewership by 50%.
- **Fostered a collaborative, team-oriented environment**, mentoring new members and ensuring effective communication across technical and non-technical stakeholders.

ADDITIONAL

- **Technical Skills:** Java, Python, JavaScript, FastAPI, React, Expo, Supabase, Express, NEAT, SQL
- **Work Eligibility:** Eligible to work in the U.S. with no restrictions